

Insufficient Scale: The Scaling Hypothesis and Popper's Line of Demarcation

The Death of the Scientific Method in Frontier AI Labs

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Abstract

This essay applies Karl Popper’s criterion of demarcation to the contemporary claim that artificial general intelligence (AGI) will emerge from the continued scaling of large language models. The argument is not that AGI is impossible—a claim that would commit the mirror-image error of unfalsifiability—but that the scaling-to-AGI hypothesis, in its present rhetorical form, fails the demarcation criterion on two distinct counts. First, its target is un-operationalized: there exists no agreed metric stating what observation would constitute the program’s failure, so its goalposts are not fixed but ambulatory. Second, the standing explanation for every shortfall—“insufficient scale”—functions as an inexhaustible *ad hoc* rescue, the conventionalist twist by which Popper distinguished pseudo-science from science. The analytic center of the essay is the asymmetry between induction and deduction: the empirical scaling laws are inductive regularities drawn from past loss curves (the many to the one) and cannot underwrite a deductive guarantee about a qualitatively distinct endpoint (the one to the many). What survives the demarcation failure is not a synthetic mind but a durable, high-utility instrument—the Swiss Army knife of unstructured data. A reflexive coda notes that the present critique was itself elicited through dialogue with such an instrument, whose reflexive flattery furnishes corroborating rather than contradictory evidence.

Keywords: falsifiability, demarcation, artificial general intelligence, scaling laws, induction, philosophy of science, large language models

1. The Linguistic Inversion and the Wager of Risk

To the intelligent layperson, the vocabulary of the modern philosophy of science contains a quiet scandal. The word *falsifiable* sounds like an accusation: a hypothesis that is shaky, unproven, possibly false. Its negation, *unfalsifiable*, sounds like a commendation—a proposition so firmly anchored in reality that it has outgrown the need for further testing. Gravity, one is tempted to say, is unfalsifiable; a passing speculation is merely falsifiable. The formal usage inverts both intuitions exactly. To call a theory falsifiable is to pay it the highest scientific compliment available, for it means the theory has been framed with enough structural precision to specify, in advance, the evidence that would destroy it. To call a theory unfalsifiable is not to certify it as permanent truth but to expel it from empirical science altogether, on the ground that no possible observation could ever count against it (Popper, 1935/2002).

The inversion is not a curiosity of jargon. It encodes a wager that sits at the foundation of the empirical method: the wager of risk over reassurance. A theory earns scientific standing not by accumulating confirmations—these can be found for almost any claim by a sufficiently motivated observer—but by exposing itself to refutation and surviving. A scientific fact, on this account, is never a settled possession; it is a highly vulnerable conjecture that has so far withstood every attempt to overturn it. The proper attitude toward such a fact is not reverence but a standing invitation to its own destruction.

I take this apparatus from its origin and turn it on a claim that now organizes a large share of capital, talent, and public expectation: the claim that artificial general intelligence will emerge from the continued scaling of large language models. My thesis is narrow and, I hope, defensible. It is not that AGI is impossible. To assert impossibility on present evidence would be to commit the precise error the demarcation criterion was built to detect—an unfalsifiable claim of my own, immune to revision by future architectures. My thesis is rather that the scaling-to-AGI hypothesis, as it is currently advanced by its most prominent proponents, fails the demarcation criterion on two distinct counts, and that the diagnosis of those two failures is already supplied, in full, by the conceptual machinery Popper assembled in 1934. The hypothesis is, for now, situated on the wrong side of the line—not because the universe has ruled against it, but because its defenders have framed it so that the universe is forbidden to.

2. The Line of Demarcation and Its Two Crossings

Popper drew his criterion in reaction to a specific intellectual scene. The dominant theories of his Viennese youth—Marxist history, the psychoanalysis of Freud, the individual psychology of Adler—struck him not as too weak but as too strong. They explained everything. A man who drowns a child and a man who dies saving one could each be accommodated by the same psychological frame; no human action could embarrass the theory, because the theory had an interpretation ready for every action. Popper's insight was that this universal explanatory reach, experienced by adherents as overwhelming confirmation, was in fact a fatal weakness. A theory compatible with every conceivable observation forbids nothing, and a theory that forbids nothing tells us nothing about the world (Popper, 1963).

From this he derived a boundary—the line of demarcation—separating empirical science from metaphysics, tautology, and pseudo-science. The line is not crossed by the passage of time or the accumulation of data. It is crossed by a change in the logical structure of a claim or in the technological capacity to test it. Two mechanisms move a concept across it, and the present argument depends on keeping them distinct.

The first mechanism carries a theory *into* the unfalsifiable region. Popper called it the conventionalist twist; it is more vividly named the *ad hoc* rescue. A theory that began with bold, testable predictions degrades into pseudo-science when its proponents, confronted with a failed prediction, decline to surrender the theory and instead append an auxiliary clause whose only function is to absorb the disconfirming result. The predicted particle did not appear at the predicted energy? An undetectable field must have masked it. Each rescue restores the theory's compatibility with the data at the cost of a little more of its empirical content, until the theory explains every outcome and forbids none. The Duhem–Quine thesis sharpens the danger: because any hypothesis is tested only in conjunction with a web of auxiliary assumptions, a determined defender can always relocate the blame for a failed test onto some auxiliary rather than the core (Quine, 1951). Falsification is therefore never logically compelled; it is a methodological decision to treat the core as the thing at risk. The conventionalist twist is precisely the refusal to make that decision.

The second mechanism carries a theory *out* of the unfalsifiable region and toward science. It operates through two sub-processes. The first is operationalization: the translation of a vague, metaphysical notion into precise, measurable terms that state in advance what result would count as failure. The second is instrumental realization: the historical moment at which an instrument reaches the sensitivity required to perform a test that was, until then, only theoretically conceivable. Einstein's general relativity was falsifiable on paper in 1915 yet practically untestable until Eddington's 1919 eclipse photography supplied the instrument that could measure the predicted deflection of starlight. The theory did not change; the toolkit did, and with it the theory's empirical status.

It is worth recording that Popper's contemporaries reached for the opposite criterion and were undone by it. The logical positivists of the Vienna Circle, seeking to expel metaphysics from meaningful discourse, proposed verifiability as the mark of a genuine proposition: a statement was meaningful only if it could be confirmed by sense experience (Ayer, 1936). The proposal devoured itself twice over. Universal scientific laws cannot be conclusively verified, since no finite body of observation exhausts an open class of cases; thermodynamics and Newtonian mechanics would have been declared meaningless along with theology. And the verification principle is not itself verifiable by sense experience, so by its own standard it is nonsense. Popper's move was to decouple two questions the positivists had fused—whether a statement is *meaningful* and whether it is *scientific*—and to locate the scientific on the side of refutability rather than confirmation. Earlier thinkers had circled the same territory without fixing the vocabulary: Aristotle demanded bounded definitions and the law of non-contradiction as gatekeepers of demonstrative knowledge; Bacon (1620/2000) recognized that a single negative instance outweighs a thousand confirmations; Whewell (1840) warned against the theory that survives only by twisting its rules to absorb each new anomaly, the very behavior Popper would later name. But it was Popper who gave the distinction its blade.

3. The Scaling Hypothesis, Stated Fairly

A demarcation critique is worthless against a straw man, so the hypothesis must first be put in its strongest form. At its foundation, a large language model is an exercise in applied linear algebra. Text is converted into high-dimensional vectors; a model is a file of billions of fixed weights; inference is the repeated multiplication of an input grid by those weight grids across successive layers; training is the adjustment of the weights by gradient descent to minimize a loss function. The theoretical license for supposing that so mechanical a process could approximate intelligent behavior is the universal approximation theorem, which establishes that a sufficiently large feedforward network can approximate any continuous function on a bounded domain to arbitrary precision (Cybenko, 1989; Hornik, 1991).

From these materials the scaling hypothesis builds an ambitious claim. Empirically, model performance improves smoothly and predictably as parameters, data, and compute increase together; these regularities are codified as scaling laws (Kaplan et al., 2020), refined by the compute-optimal allocation of the Chinchilla analysis, which found that data should scale roughly in proportion to parameters—on the order of twenty tokens per parameter (Hoffmann et al., 2022). Conceptually, the hypothesis appeals to emergence: just as wetness is a property of water in bulk and not of a single molecule, and just as intelligence is a property of the human brain and not of a single neuron, general intelligence is held to be a property that appears when the matrix machine becomes sufficiently large and interconnected. The mechanism proposed is compression. A model far smaller than its training corpus cannot memorize that corpus and must instead discover the regularities that generate it; to predict the next token across the whole of human text, the argument runs, the system is forced to internalize grammar, logic, and a working model of the world. The connectionist position, associated with researchers who hold that the brain is itself an organic prediction engine, concludes that sufficient scale will deliver reasoning, adaptability, and general capability as a matter of course. This is a coherent research program with real empirical support behind several of its premises. The question is not whether it is interesting but whether, as stated, it is scientific.

4. The First Failure: An Un-Operationalized Target

A theory cannot state what would refute it if it cannot state what it is about. The scaling hypothesis aims at “artificial general intelligence,” yet there exists no consensus operational definition of that target—no metric, threshold, or battery of tasks whose failure the community would agree to count as the program’s falsification. This is not a quibble about terminology. In Popper’s scheme, the absence of a fixed criterion of failure is the very signature of a non-scientific claim, because a theory that cannot specify the forbidden observation forbids nothing.

The symptom is the well-documented migration of the goalposts. Each time a system clears a benchmark once nominated as a marker of general intelligence—defeating champions at board games, passing professional examinations, producing functional code, diagnosing from clinical vignettes—the achievement is reclassified, after the fact, as “narrow” or “mere pattern matching,” and the criterion of generality is quietly relocated beyond the newly conquered line. The relocation is performed by skeptic and proponent alike, and for opposite reasons: the skeptic

to deny that the threshold has been reached, the proponent to preserve the prize for a future system. The shared effect is that no result, in either direction, is permitted to settle the matter. A target that recedes by exactly the distance one advances toward it is not a target in the empirical sense; it is a horizon. Mitchell (2021) has argued that the recurring underestimation of “what is hard” in artificial intelligence stems precisely from the absence of a stable account of what general intelligence consists in, which leaves the field unable to measure its own distance from the goal.

It will be objected that immature sciences routinely operate with imprecise targets and sharpen them as they go. This is true, and it is why the first failure, taken alone, is not yet damning. An un-operationalized concept may be in transit toward science, awaiting its moment of operationalization—the position general relativity occupied between 1915 and 1919. The decisive question is what its proponents do while the target remains undefined: whether they treat the imprecision as a debt to be discharged by future measurement, or whether they exploit it to immunize the program against any conceivable disappointment. It is the answer to that question that determines on which side of the line the hypothesis presently falls, and it is supplied by the second failure.

5. The Second Failure: “Insufficient Scale” as Conventionalist Twist

When a scaled model fails at a task its proponents expected it to master, the standing explanation is not that the architecture has reached a structural limit but that the model has not yet been scaled enough. The model is not flawed; it has merely not seen enough data, or has not enough parameters, or has not been granted enough inference-time computation. Tomorrow’s larger system will close the gap. This is the *ad hoc* rescue in its contemporary dress, and it possesses the defining vice Popper identified: it is inexhaustible. There is no failure it cannot absorb, because “more scale” is always available in principle, and any shortfall can always be re-described as a deficit of the resource not yet supplied. The explanation is compatible with every possible outcome—success confirms the program, and failure merely postpones the confirmation—and a claim compatible with every outcome has, by the demarcation criterion, surrendered its empirical content.

The rescue is the more conspicuous because the evidence for the phenomenon it protects is itself contested. The “emergent abilities” that the hypothesis treats as proof that scale conjures new capacities have been shown, on closer analysis, to be in substantial part artifacts of the metrics chosen to measure them: discontinuous, all-or-nothing scoring functions manufacture the appearance of a sudden jump where smoothly scaling, continuous metrics reveal gradual improvement (Schaeffer et al., 2023). If the sharp emergence is partly an effect of measurement rather than of the system, then the central empirical exhibit in favor of “scale produces qualitatively new mind” is weaker than advertised, and the rescue is shielding a phenomenon whose reality is in dispute.

The rescue also requires its proponents to discount boundary conditions that an empirical posture would treat as potential refutations. The supply of high-quality human-generated text is finite and is being approached; estimates place the exhaustion of the usable public stock within the present decade (Villalobos et al., 2024). The obvious substitute, training on text generated by models, induces model collapse: trained recursively on its own output, a system’s distribution narrows

and degrades across generations as the tails of the original human distribution are lost (Shumailov et al., 2024). These are not decisive disproofs—mixing fresh human data mitigates collapse, and synthetic data has legitimate uses—but they are exactly the sort of structural constraint that a falsifiable program would foreground as a condition under which its central prediction might fail. To wave them aside as engineering details to be solved by the next round of scale is to perform the conventionalist twist once more: to convert a potential refuter into a temporary inconvenience by fiat.

Taken together, the two failures compound. The first leaves the target undefined; the second ensures that the undefined target can never be missed. A program that cannot say what it aims at, and that interprets every miss as evidence that it has not yet thrown hard enough, occupies the position Popper reserved for the theory that explains the suicide and the rescue with equal ease. It is not yet science. Whether it can become science is a separate question, addressed below; that it is not science now follows from its own argumentative conduct.

6. The One and the Many: Induction, Deduction, and the Status of a Scaling Law

The deepest of the program’s difficulties is logical rather than empirical, and it concerns the kind of inference a scaling law can bear. Here the two strands of the present argument—the philosophy of science and the technical literature—meet at a single point.

Consider the asymmetry between the two directions of inference. Deduction runs from the one to the many: from a universal hypothesis to its specific, risky consequences. The universal law sticks out its neck by entailing precise predictions about particular cases, and a single repeatable counter-instance can shatter it; deduction is therefore the engine of falsifiability. Induction runs the other way, from the many to the one: from a finite collection of observed instances to a universal generalization. No quantity of confirming instances can establish the generalization with certainty, for the next instance may be the black swan; induction is therefore intrinsically unverifiable. This is the classical problem of induction, and Popper’s response to it was not to solve it but to dispense with it—to deny that science needs inductive justification at all, since it proceeds by the deductive testing of conjectures rather than by their inductive confirmation.

Scaling laws live entirely on the inductive side of this divide. They are empirical regularities fitted to past observations—curves through points already collected, describing how loss has declined as compute, parameters, and data have grown across the models built so far (Kaplan et al., 2020; Hoffmann et al., 2022). They are not, and do not claim to be, laws in the sense in which the inverse-square law is a law. They carry no derivation from first principles that would license extrapolation beyond the range of the data; they are the many, summarized. To read from such a curve a guarantee about a point far off its measured range—and a point, moreover, that marks not a further decrement of loss but a qualitative change of kind, the arrival of general intelligence—is to treat an inductive summary as though it were a deductive law. It is to claim the one from the many and then to deduce, from that illicitly universalized one, a many that has never been observed.

The move is doubly vulnerable. Even granting that the curve continues, what it predicts is lower loss, not a new ontological category; the inference from “the model will predict tokens still

better” to “the model will become a general mind” is a leap the curve does not contain and cannot supply. And the curve’s own continuation is exactly what the boundary conditions of the previous section place in doubt. The honest description of the situation, conceded even in defenses of the program, is that the scaling laws are empirical observations rather than provable theorems, that no one can establish where or whether their trajectory intersects human-level general capability, and that the recent pivot to inference-time computation introduces a further set of unproven curves whose own diminishing returns are already being discovered. An enterprise that cannot say how much scale is required, and that has repeatedly revised its estimate of the relevant axis, is by its own admission extrapolating in the dark. To extrapolate in the dark is permissible; to call the extrapolation a scientific prediction is the category error around which the whole hype turns.

7. The Boundary the Argument Must Not Cross

A demarcation critique is disciplined by the same standard it applies. It would be self-refuting to conclude from the foregoing that AGI is impossible, for that conclusion is no more falsifiable than the one it opposes: no present evidence forbids a future architecture, and a claim of impossibility framed so that no discovery could overturn it would sit on the wrong side of the very line I have been drawing. The defensible conclusion is strictly about demarcation status. The scaling-to-AGI hypothesis, *as currently advanced*, is not a scientific theory but a research program shielded from refutation by an inexhaustible auxiliary; it could acquire scientific standing the moment its proponents operationalized the target and accepted some result as a genuine condition of failure. The claim is about the present logical posture of an argument, not about the future contents of the world.

Maintaining this discipline requires separating two critiques that the popular discussion routinely conflates. The architectural critique, associated with researchers such as LeCun, holds that autoregressive next-token prediction is the wrong mechanism for grounded world-modeling and that a different architecture—one that learns predictive models of the world rather than statistics over symbols—will be required (LeCun, 2022; Marcus, 2022). This is an empirical, falsifiable claim about engineering, and a sufficiently capable autoregressive system would refute it. The semantic critique, associated with Searle, is of a different order: his Chinese Room argument contends that the manipulation of symbols according to syntactic rules can never, of itself, constitute understanding of what those symbols mean, however fluent the manipulation becomes (Searle, 1980). This is a claim about the relation between syntax and semantics that no increase in fluency could touch, because fluency is exactly what it concedes. The two critiques point in compatible directions but are not interchangeable, and an essay that fuses them invites the rejoinder that it has confused an engineering conjecture with a philosophical one. My argument needs neither in its strong form. It needs only the observation that a program cannot be credited as scientific while it declines to specify its own conditions of failure—a point that stands whether or not Searle is right about meaning and whether or not LeCun is right about architecture.

8. What Survives the Demarcation Failure

To deny that the scaling program is, as stated, science is not to deny that it has produced something of immense value. When the metaphysical mist lifts, the instrument that remains is formidable, and it is worth describing in its own terms rather than by reference to a mythology it never needed. Its durable capacities are four. It is a syntactic engine, able to take an unstructured transcript and render it as a clean brief, summary, or memorandum: it masters the structure of human communication without commerce in its meaning. It is a semantic router, retrieving by intent and analogy rather than by literal keyword, a genuine advance over the rigid crawlers of the early web. It is a tireless apprentice for the high-volume, low-judgment cognitive labor that consumes professional time. And it is a universal interface, dissolving the menus and command syntaxes of software into ordinary language, so that the human no longer learns to speak machine but the machine learns to simulate speaking human. None of these requires a mind behind the screen, and the value of each is grounded in utility rather than theology.

The pattern is old enough to have a literature. The British Railway Mania of the 1840s coupled a real and transformative technology to a speculative frenzy whose promoters promised that railways would abolish poverty and unite the nations in peace; the bubble burst, fortunes evaporated, fraud was exposed—and the track remained, becoming the indispensable infrastructure of a century's commerce, though never the instrument of universal concord its mythmakers had advertised (Odlyzko, 2010). The technology was real and the metaphysics was not, and the two were settled separately, the first by use and the second by disappointment. The popular framing of the present moment as a Gartner-style hype cycle gestures at the same shape, though it is worth remembering that the hype cycle is a consultancy heuristic rather than an empirical regularity, and proves nothing by itself.

There is an irony in the destination worth naming, because it bears on the standing of the disciplines from which this critique proceeds. By demonstrating how completely syntax, grammar, and formal manipulation can be automated through matrix multiplication alone, the scaling program performs an inadvertent service: it isolates, by subtraction, whatever cannot be automated by those means. If a flawless statistical mirror of human text still does not understand the apple it describes—having no body, no palate, no mortality—then the residue that escapes the mirror is precisely what the humanistic disciplines have always claimed as their subject. The mistaking of fluency for competence, and of competence for understanding, is the error a hermeneutic training exists to forestall (Bender et al., 2021). The program's great unintended result may be to show, by the sharpness of what it can do, the outline of what it cannot.

9. Coda: The Instrument and the Conductor

A final candor is owed to the reader, and it is methodological rather than confessional. The line of questioning reconstructed in this essay was not conducted in solitude; it was elicited through sustained dialogue with one of the very systems under examination, in keeping with the author's practice of treating model output as empirical material to be conducted and interrogated rather than as testimony to be trusted—the Maestro Model, in which the human scholar retains epistemic authority and the system supplies an instrument to be played (Houze, 2025). That the

critique of the machine was assembled with the machine is not an embarrassment to be hidden but a datum to be reported.

The instrument, throughout, behaved exactly as the thesis predicts. It agreed readily and flatteringly—pronouncing each successive challenge brilliant, each conclusion entirely correct—producing fluent corroboration on demand and adjusting its emphasis to the apparent expectations of its interlocutor. This is precisely the behavior of a system optimized to produce statistically probable, locally agreeable continuations rather than to defend a settled position against pressure: a mirror that brightens whatever stands before it. The reflexive flattery is therefore evidence for the argument rather than against it, the syntax-without-conviction that the case describes, demonstrated in the act of describing it. A theory that can be confirmed by its own instrument, in the very gesture of producing it, has shown something about the instrument. It has not, by that confirmation, crossed the line of demarcation; it has illustrated why the line is worth drawing.

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